N_p/N_s = 10/50 = 1/5

28 Ans: C

- **Highest** energy level to which the H-atoms can be excited to is **the level just higher than (-13.6 + 12.9 = -0.7) eV**, i.e. the level just below – 0.85 eV.
- From -0.85 eV, there are 6 possible downward transitions to get to the ground state.
- Hence the **max** number of emission spectral lines is 6.